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Patient-specific myocardial infarction risk thresholds from AI-enabled coronary plaque analysis

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Abstract

Background: Plaque quantification from coronary computed tomography angiography (CTA) has emerged as a valuable predictor of cardiovascular risk. Deep learning (DL) can provide

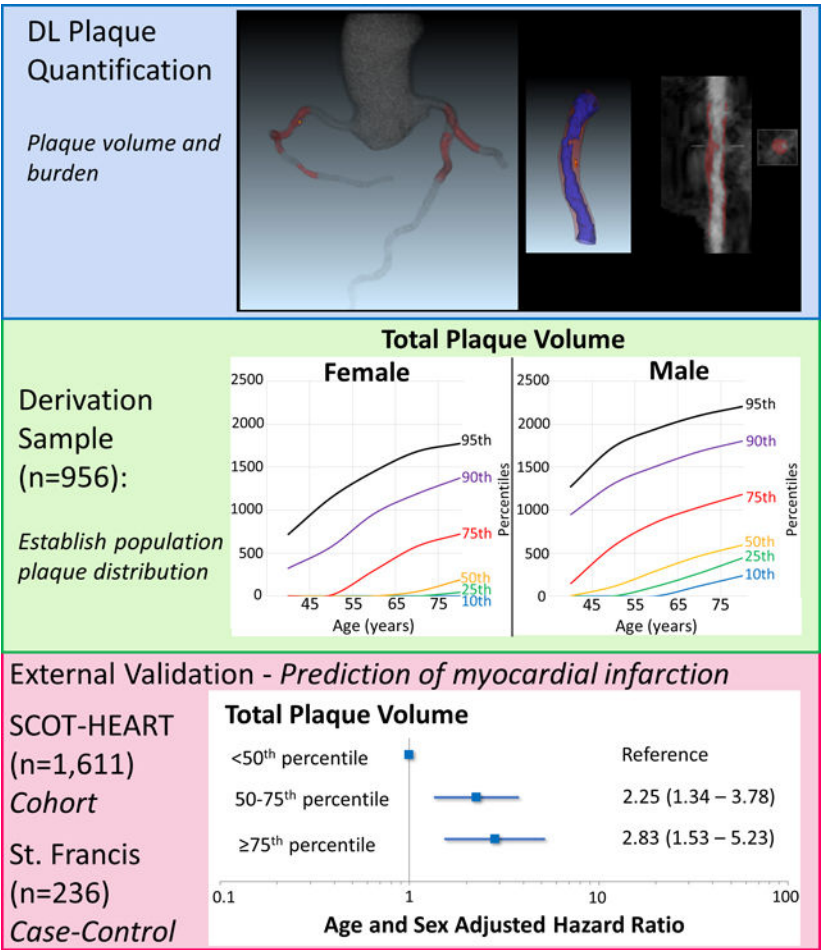
automated quantification of coronary plaque from CTA. We determined per-patient age and sex-specific distributions of DL-based plaque measurements and further evaluated their risk prediction for myocardial infarction in external samples.

Methods: In this international, multicenter study of 2803 patients, a previously validated DL system was used to quantify coronary plaque from CTA. Age and sex-specific distributions of coronary plaque volume were determined from 956 patients undergoing CTA for stable coronary artery disease from 5 cohorts. Multicenter external samples were used to evaluate associations between coronary plaque percentiles and myocardial infarction.

Results: Quantitative DL plaque volumes increased with age and were higher in male patients. In the combined external sample (n=1,847), patients in the $\geq 75^{\text{th}}$ percentile of total plaque volume (unadjusted hazard ratio 2.65, 95% confidence interval 1.47–4.78, $p=0.001$) were at increased risk of myocardial infarction compared to patients below the 50th percentile. Similar relationships were seen for most plaque volumes and persisted in multivariable analyses adjusting for clinical characteristics, coronary artery calcium, stenosis and plaque volume, with adjusted hazard ratios ranging from 2.38 to 2.50 for patients in the $\geq 75^{\text{th}}$ percentile of total plaque volume.

Conclusions: Per-patient age and sex-specific distributions for deep learning-based coronary plaque volumes are strongly predictive of myocardial infarction, with the highest risk seen in patients with coronary plaque volumes in the $\geq 75^{\text{th}}$ percentile.

Graphical Abstract



Keywords

Deep learning; coronary plaque; risk prediction; coronary CT Angiography; sex-specific analysis; myocardial infarction

Subject Terms:

Computed Tomography (CT)

INTRODUCTION

Coronary artery disease (CAD) is a leading cause of morbidity and mortality with several non-invasive imaging modalities available for disease diagnosis and risk prediction¹. Given its high diagnostic accuracy, coronary computed tomography angiography (CTA) was given a class 1 recommendation as an initial test for diagnosing CAD in symptomatic patients¹. CTA can also improve estimation of cardiovascular risk^{2, 3}, potentially allowing physicians to more accurately target medical therapies^{4, 5}.

There are several different metrics available for the estimation of cardiovascular risk from cardiac CT. Coronary artery calcium (CAC) continues to be used extensively as it provides an overall assessment of the extent of atherosclerosis and has been shown to be a robust predictor of cardiovascular risk⁶⁻⁸. Visual assessment of the extent of nonobstructive plaque and identification of high-risk plaque features carries prognostic value and can be also performed during routine clinical reporting^{9, 10}. Quantitative plaque analysis using semi-automated software provides a more robust assessment of plaque burden and morphology¹¹⁻¹³. These volumetric measurements include total plaque (TP) and its components: non-calcified plaque (NCP), low-attenuation plaque (LAP), and calcified plaque (CP)¹¹⁻¹³. TP volume measurements improve prediction of cardiac death or acute coronary syndrome when added to clinical risk and CAD severity¹³. However, current methods for quantification of these parameters are time-consuming and require high levels of expertise, limiting their clinical adoption.

We developed a convolutional long short-term memory (ConvLSTM) system for fully automated quantification of coronary plaque volumes from CTA¹⁴. This deep-learning (DL) system demonstrated excellent agreement with plaque segmentation by expert readers (intra-class correlation 0.810 to 0.964). Importantly, it requires only a fraction of the expert analysis time (3.8 minutes vs. 25.7 minutes on average), and thus has potential for routine clinical use¹⁴.

In this study, we determined population distributions for DL coronary plaque volume and burden using a large international multicenter cohort of patients. Importantly, we then externally validated the prognostic significance of plaque volume percentiles for fatal or non-fatal myocardial infarction in patients from the Scottish Computed Tomography of the Heart (SCOT-HEART) trial and the St. Francis Hospital and Heart Center.

METHODS

Study Cohorts

This study was performed in a large international multicenter cohort of 2803 patients and was performed using derivation and two external testing samples. The data that support the findings of this study are available from the corresponding author upon reasonable request. The derivation sample included 965 patients who underwent CTA for a stable CAD indication with available information regarding age, sex, and DL-based coronary plaque volumes¹⁵. We excluded patients with CTA scans of poor image quality that were deemed uninterpretable by expert readers (n=9), leaving 956 patients in the analysis. Additional details for each of the cohorts in the derivation sample are available in Table S1. The derivation sample was used to establish population distributions for coronary plaque volumes and included both patients with and without symptoms.

We included two external samples, one cohort and one case-control, to provide validation of the prognostic importance of plaque percentiles. The first external testing sample was comprised of 1,611 patients from the prospective SCOT-HEART randomized controlled trial with available information regarding age, sex, DL-based coronary plaque volumes and clinical follow-up for fatal or non-fatal myocardial infarction¹⁶. SCOT-HEART included

patients aged 18–75 with stable chest pain who were referred to a cardiology clinic for evaluation and underwent coronary CTA in the CTA arm of the SCOT-HEART trial (16). We relied on CAD stenosis interpretation from the SCOT-HEART trial, in which coronaries were visually assessed on a 15-segment basis by trained observers with SCCT Level III certification in cardiac CT, as described previously¹⁷. In the present study, obstructive CAD was defined as left main stenosis $\geq 50\%$ or $\geq 70\%$ in other segments.

The second external testing sample was derived from selected patients without known CAD who underwent clinically indicated coronary CTA from 2007–2018 at St. Francis Hospital and Heart Center, Roslyn, New York. Within this case-control sample, patients who experienced myocardial infarction during follow-up (49 patients, median follow-up 2.5 years) were matched to controls based on age, sex, body mass index, and CTA acquisition tube voltage. Overall, 236 patients (49 cases, 187 matched controls) were included with symptoms of either chest pain (125, 53%) or dyspnea (111, 47%). This study complies with the Declaration of Helsinki and was approved by the institutional review boards from participating sites. Patients either provided written informed consent or a waiver of written consent was approved by the local institutional review board based on site-specific protocols.

DL Architecture

The DL architecture is a hierarchical ConvLSTM network developed to segment coronary arteries, as described in detail previously¹⁴, and is summarized in the Supplemental Methods section. The model is integrated in an onsite plaque analysis software package (Autoplaque v3.0; Cedars-Sinai Medical Center), which has been cleared by the Food and Drug Administration. The DL network was trained using expert annotated coronary segments ≥ 2.0 mm within the 18-segment coronary model¹⁴. Per-patient plaque volumes are quantified from segmented non-calcified plaque, lumen, and calcified plaque over the coronary tree. Following plaque segmentation, adaptive scan-specific Hounsfield unit thresholds were applied to define non-calcified and calcified plaque constituents as described previously¹⁸, which corrects for differences related to scan parameters. Low-attenuation plaque was defined using a fixed threshold of < 30 Hounsfield Units. Plaque burden is calculated as plaque volume/analyzed segment volume. An example of DL plaque quantification results in a patient who experience myocardial infarction is shown in Figure 1.

Establishing Population Distributions

Distributions for coronary plaque volume were derived exclusively in the derivation sample. We followed the methodology utilized in the Multi-Ethnic Study of Atherosclerosis (MESA) for deriving age and sex-specific CAC distributions¹⁹. First, a mixed-effects regression model was developed to predict coronary plaque volumes or plaque burden from age, sex, and an interaction between the two variables. In this model, cohort was included as a grouping feature to account for heterogeneity between cohorts²⁰. Next, differences between actual plaque volumes and expected plaque volumes were determined and assigned raw percentiles of plaque volume for each patient. Lastly, within each sex and age category, the raw percentiles were re-categorized to be a function of non-zero values as previously described¹⁹. These final percentiles were utilized to establish expected population

distributions. This is summarized in Figure S1. This approach creates a pooled residual distribution which improves the precision for percentile estimates compared to simply calculating age and sex sub-groups (particularly given the limited number of patients for some age and sex categories)¹⁹. The distribution of patients according to percentile of TP volume as a function of age and sex categories are shown in Table S2.

Clinical Outcomes

The primary outcome for this study was fatal or non-fatal myocardial infarction. Outcome data were obtained electronically from the electronic Data Research and Innovation Services of the National Health Services Scotland and were verified by review of electronic health records when required¹⁶. Follow-up was censored on January 1, 2018 for patients without an event¹⁶. In the St. Francis sample, events were ascertained through electronic medical records and adjudicated by an experienced physician.

Statistical analysis

Continuous variables are summarized as median and interquartile range and compared using the Wilcoxon rank sum test since they were not normally distributed. Categorical variables are summarized as number (frequency) and compared using chi-square tests.

Cox proportional hazards models were used to assess associations between plaque volume or burden percentile with the primary outcome using unadjusted analyses and adjusted analyses. We combined the external samples to evaluate models accounting clinical and CAC, stenosis, and plaque volumes in sequential models incorporating all variables. In analyses of the combined external samples, shared frailty models were utilized to account for heterogeneity between samples²¹. The clinical and CAC multivariable model included age, sex, risk factors (hypertension, diabetes, dyslipidemia, smoking, family history), and CAC score in addition to the plaque percentiles. The stenosis model included the variables from the clinical and CAC model and the presence of obstructive CAD in addition to the plaque percentiles. The plaque model included the variables from the stenosis model and plaque volumes as a continuous variable in addition to the plaque percentiles.

Improvement in model fit was assessed using the likelihood-ratio chi-square test. We evaluated Harrell's C statistic for models without TP volume percentiles compared to models with TP volume percentiles. Lastly, we evaluated continuous net reclassification index for the addition of plaque volume and burden percentiles to the three multivariable models. All tests were two-sided with $p\text{-value} < 0.05$ consider significant. All calculations were performed using Stata version 14.0 (StataCorp, College Station, TX, USA).

RESULTS

Patient Characteristics

In total, 956 patients from 5 cohorts were included in the derivation sample. Details of the derivation and external validation cohorts are shown in Table 1. Patients in the derivation sample had a higher prevalence of hypertension (54.5% vs 36.1%, $p < 0.001$) and lower prevalence of smoking (9.0% vs 18.5%, $p < 0.001$) compared to patients from SCOT-HEART.

The derivation cohorts included a higher proportion of patients with dyslipidemia (69.9% vs 58.0%, $p=0.003$) compared to patients from St. Francis.

Distribution of Deep-Learning Plaque Quantification

TP volumes were higher in male patients and increased with age (Figure 2). The 75th percentile ranged from 0 to 720 mm³ in female patients and 147 to 1180 mm³ in male patients. The 95th percentile ranged from 720 to 1773 mm³ in female patients and 1273 to 2203 mm³ in male patients. NCP volumes were higher in male patients compared to female patients ($p<0.001$; Figure S2). With increasing age, there was an increase in NCP volumes, with the 75th percentile ranging from 0 to 512 mm³ in female patients and 172 to 1042 mm³ in male patients. LAP volume, CP volume, NCP burden, and LAP burden were also higher in male patients and increased with age, with distributions shown in Figures S3-S6.

External Validation

In the SCOT-HEART sample, during median follow-up of 4.8 (interquartile range 4.1 – 5.8) years, 41 patients experienced fatal ($n=3$) or non-fatal ($n=38$) myocardial infarction. Of those events, 20 occurred in patients with obstructive CAD and 21 occurred in patients without obstructive CAD. By contrast, 24 events occurred in patients in the $\geq 75^{\text{th}}$ percentile for TP and 13 in patients in the 50–75th percentile (Figure S7). Kaplan-Meier survival curves according to percentiles of LAP volume, NCP volume, CP volume, NCP burden, and LAP burden are shown in Figures S8-S12.

During median follow-up 2.5 years (interquartile range 1.0 – 4.8), 49 patients experienced non-fatal myocardial infarction in the St. Francis Hospital and Heart Center cohort. There was no difference in age (median 65 vs 67, $p=0.923$), sex (61% vs 59%, $p=0.870$), or presence of obstructive CAD (14.3% vs 11.2%, $p=0.619$) in patients who experienced myocardial infarction compared to matched control patients. However, TP volume was higher in patients who experienced myocardial infarction (median 175 versus 108 mm³, $p=0.004$).

Age and sex adjusted associations with fatal or non-fatal myocardial infarction in the combined external samples are shown in Figure 3. There was an increased risk of fatal or non-fatal myocardial infarction with increasing plaque volume, regardless of the plaque type, with age and sex adjusted HR ranging from 1.90 to 3.33 for patients in the $\geq 75^{\text{th}}$ percentile. Associations between plaque volume and burden percentiles in multivariable models are shown in Table 2. Patients in $\geq 75^{\text{th}}$ percentile for all plaque volumes were at significantly increased risk of fatal or non-fatal myocardial infarction compared to patients below the 50th percentile when models were adjusted for age, sex, risk factors, and CAC score, (adjusted HR 1.72 to 2.78). After further adjusting for obstructive CAD, TP, NCP, and LAP volume percentiles were associated with myocardial infarction (adjusted HR 2.39 to 3.07). Similar results were seen after further adjusting for plaque volume as a continuous variable.

The multivariable models for TP volume percentiles are shown in Table S3. Notably, the presence of obstructive CAD was not associated with myocardial infarction in multivariable models. Model fit was significantly improved by including plaque volume percentile in most

analyses (Table 2). Evaluation of the net-reclassification analysis is shown in Table S4, with NRI ranging from 0.107 to 0.265 for plaque volume percentiles.

DISCUSSION

In this study, we established population distributions of coronary CTA-based DL plaque volumes in a large, multicenter, international cohort. Population distributions of plaque volumes (NCP, LAP, CP, and TP) increased with age and were higher in male patients. To our knowledge, our study is the first to show that higher age and sex-specific percentile of plaque volume was associated with an increased risk of fatal or non-fatal myocardial infarction in two external samples, including the landmark SCOT-HEART trial. Patients in the $\geq 75^{\text{th}}$ percentile had a risk of myocardial infarction up to 3 times higher than patients below the 50^{th} percentile in adjusted analyses. Lastly, we show that plaque volume percentiles could be used to improve model fit and patient classification when added to predictive models incorporating clinical risk factors and CAC, CTA stenosis, or plaque volume as a continuous variable. Future studies should evaluate whether high risk patients, based on population distributions of DL coronary plaque volume, derive clinical benefit from more intensive medical therapy.

With the emergence of CTA plaque quantification as a potential clinical tool, there is a clinical need to understand population distributions and their associated risk of events. As expected, population distributions of plaque volumes increased with age and were higher in male patients. Tzimas et al. recently presented a similar analysis using another plaque quantification software, showing similar expected population distributions for TP volume²². For example, the median TP volume for a 50-year-old male in their analysis was $\sim 100 \text{ mm}^3$ compared to 109 mm^3 in our analysis and the 90^{th} percentile for an 80-year-old woman was $\sim 1100 \text{ mm}^3$ in their analysis compared to 1350 mm^3 in our analysis. These similarities are reassuring, particularly in light of using two different software systems and different approaches for determining population distributions. We also demonstrated the clinical utility of these percentiles for risk stratification in two external samples. Patients in higher percentile for TP, NCP, LAP, and CP volume were at higher risk of myocardial infarction in the majority of analyses. However, the presence of obstructive CAD was not associated with increased risk in the multivariable models. Additionally, over half of the events in the combined external sample occurred in patients without obstructive CAD, compared to most events occurring in patients above the 50^{th} percentile for LAP volume. It may be possible to also improve identification of plaques more likely to rupture using LAP volume quantified on a per-lesion or per-vessel basis, as demonstrated in a prior analysis of the ICONIC study²³. The strong prognostic utility of coronary plaque volumes is similar to a recent analysis of the ADVANCE registry which demonstrated that elevated total percent atheroma volume was associated with an increased risk of cardiovascular death or MI²⁴. One important difference in the present study is the use of population percentiles for group classification as opposed to an optimized cut-off from receiver-operator characteristic curve analyses. Regardless, the overall body of literature suggests that physicians could use plaque volume information to improve patient risk stratification compared to the current paradigm of using obstructive CAD. In the present study, we did not identify a consistently increased risk with higher percentile of NCP and LAP burden

(while NCP and LAP volume were associated with increased risk). This could relate to statistical power given the limited number of events or the use of population percentiles rather than optimized thresholds²⁵. Alternatively, applying age and sex-specific population thresholds may account for differences in coronary vessel volume²⁶, removing the need to normalize plaque volumes to vessel volume. Regardless, population distributions of DL plaque volumes could be referenced at the time of CTA reporting to provide this valuable information to referring physicians. A similar approach was advocated by Naoum et al. who developed age and sex-specific nomograms for segmental involvement score, demonstrating an increased risk of cardiovascular events in patients within higher percentiles²⁷. This type of comparison to other patients of a similar age and sex is an intuitive way to understand DL plaque measurements and may help motivate both physicians and patients to pursue more aggressive medical therapy in the presence of higher risk findings.

There is increasing evidence that coronary plaque volumes improve risk stratification following CTA^{28–30}. In prior work, we demonstrated that patients who suffered cardiac death had higher coronary plaque volumes compared to matched controls.³¹ Similarly, Versteyleen et al. demonstrated that per-plaque maximum volume, NCP percentage, and NCP burden were higher in 25 patients who experienced acute coronary syndrome compared to 101 random control patients³². Importantly, calcium score, luminal stenosis and plaque composition did not differ significantly between the two groups³². More recently, we demonstrated that patients from the SCOT-HEART trial with LAP burden greater than 4% were nearly 5 times more likely to experience fatal or non-fatal myocardial infarction compared to patients with lower LAP burden²⁵. In the present study, we demonstrated that plaque volume percentiles could be used to identify patients at increased risk of myocardial infarction using thresholds tested using rigorous external validation. The inclusion of plaque volume percentiles improved model fit and patient risk classification compared to models incorporating age, sex, cardiovascular risk factors, and CAC demonstrating the potential benefit from CTA over CAC scanning alone. We also showed improvement in model fit and risk classification when added to models which included presence of obstructive CAD. Overall, these results suggest that coronary plaque quantification from CTA has significant potential for improving risk stratification.

There are potential clinical applications for plaque quantification. As mentioned above, there may be a role for improving risk stratification beyond traditional considerations such as presence of obstructive CAD. In a large, multicenter study, three quarters of culprit lesions leading to acute coronary syndrome were non-obstructive and only 31% had high-risk plaque features²³. Therefore, plaque quantification may help physicians identify some high-risk patients who do not have obstructive CAD. Notably, a higher proportion of patients in the SCOT-HEART trial were experiencing chest pain compared to the derivation sample which may explain the higher number of patients categorized as being in the $\geq 75^{\text{th}}$ percentile for plaque volumes. The underlying reason may be that LAP is associated with the extent of macrophage infiltration and size of necrotic core within coronary plaques³³, markers of plaque vulnerability³⁴. While not currently a recommended indication for CCTA, physicians could potentially use this information to target more aggressive medical therapies such as PCSK9 inhibitors for those patients at highest risk. Subsequently, plaque quantification

could be used by physicians to evaluate change in coronary plaque volume composition in response to these therapies^{35, 36}.

While the clinical value of plaque quantification is becoming apparent, clinical adoption previously has been limited in part because quantitative plaque analysis using semi-automated software is time-consuming and requires a high level of expertise²⁸. However, the emergence of artificial intelligence-based automated methods for plaque quantification could reduce this barrier^{37–39}. Our current DL system can perform plaque quantification substantially faster than expert annotations even on a standard desktop computer (3.8 versus 25.7 minutes)¹⁴. Therefore, this approach could be readily integrated into clinical reporting without significantly altering workflow.

Study Limitations

Our study is not without limitations. The derivation sample included cohorts of patients with stable CAD with and without symptoms and therefore may be lower risk than some referral populations. It may be preferable to derive the population distributions in otherwise healthy patients referred for screening, but coronary CTA is rarely used in this way during clinical practice so large populations are not readily available. However, the current approach does reflect current clinical referral populations for coronary CTA at multiple international sites and the comparability to similar analyses suggests broad generalizability²². Our approach ensures that the information from all patients can inform the population percentiles in groups with smaller number of patients. However, those percentiles are reliant on assumptions inherent in the regression models and larger studies could help refine these distributions. Additionally, the populations distributions are similar to other recent analyses (suggest the results are generalizable) and the derived population distributions provided risk stratification in an external sample. The external samples included patients from the SCOT-HEART trial and a case-control sample. The latter sample was chosen given the complexity of collecting CCTA images retrospectively balanced against the need to preserve event rates. However, our results were similar when analyses were limited to patients from the SCOT-HEART trial only. We were not able to evaluate differences in plaque volumes by race or ethnicity. We know the majority of patients in SCOT-HEART (~96%) are not from ethnic minority groups⁴⁰, but have no data for the remaining samples. However, a comparison of expected plaque values by race and ethnicity as well as validation of the current percentiles in more diverse populations would be an important next step. We combined all patients below the 50th percentile for plaque volumes into a single reference category given a relatively small number of patients classified as being in the 25–50th percentile in the external samples. It is worth noting that the total number of events in the two external samples were still relatively small and may benefit from larger studies.

CONCLUSIONS

We established per-patient age and sex-specific distributions for coronary plaque volumes measured automatically with DL. Patients with larger volumes of any plaque type for age and sex were at significantly higher risk of fatal or non-fatal myocardial infarction; with the highest risk for coronary plaque volumes in the $\geq 75^{\text{th}}$ percentile.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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ABBREVIATIONS

DL	deep learning
CTA	computed tomography angiography
CAD	coronary artery disease
TP	total plaque
NCP	non-calcified plaque
CP	calcified plaque
LAP	low-attenuation plaque

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CLINICAL PERSPECTIVE

There is increasing evidence for the prognostic value of quantitative analysis of coronary plaque volume on coronary computed tomography angiography. Given the time required to quantify coronary plaque volumes manually, they will likely only be implemented clinically when quantified automatically using artificial intelligence techniques; however, physicians are not familiar with the results. As well, coronary plaque volumes are known to vary by both age and sex. In this large, multicenter, study we derived age and sex-specific distributions for coronary plaque volumes measured using artificial intelligence. We further show that patients in a higher percentile for plaque volume are more likely to experience fatal or non-fatal myocardial infarction. Physicians could potentially utilize these distributions to help explain artificial intelligence-enabled coronary plaque volumes to their patients. For example, instead of telling a patient the actual total plaque volume, it may be easier to understand the value relative to other patients of a similar age and sex. This is analogous to explaining a coronary artery calcium score within the context of age and sex-specific percentiles.

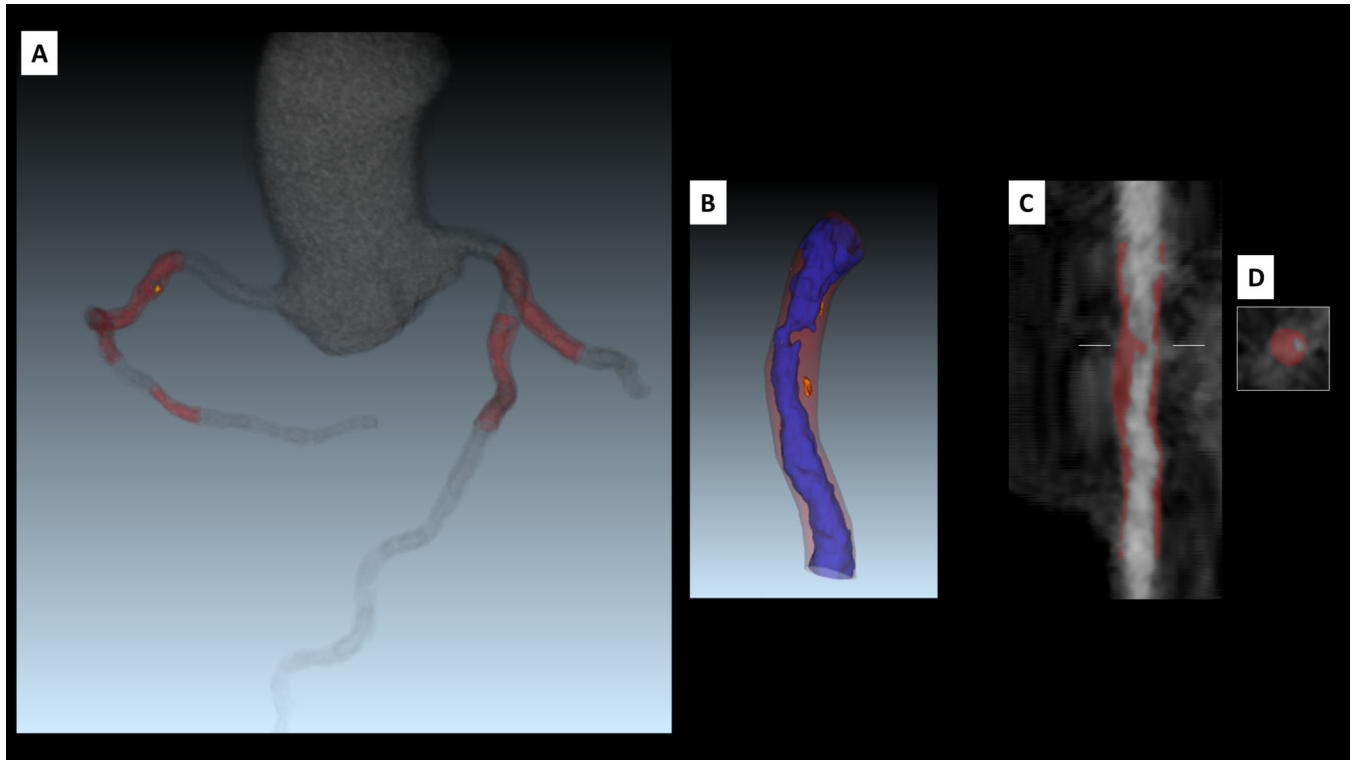


Figure 1:

Example of deep learning plaque quantification results in a patient who experienced a major adverse cardiovascular event. Panel A shows three-dimensional (3D) rendering of coronary computed tomography angiography (CTA)-based quantitative plaque analysis with red non-calcified plaque and orange low-attenuation plaque. Panel B demonstrates a 3D representation of the coronary artery lumen (blue), non-calcified plaque (red) and low-attenuation plaque (orange). The lesion is located in the right coronary artery with long-axis cross-section in Panel C, and short-axis cross-section of the lesion in Panel D.

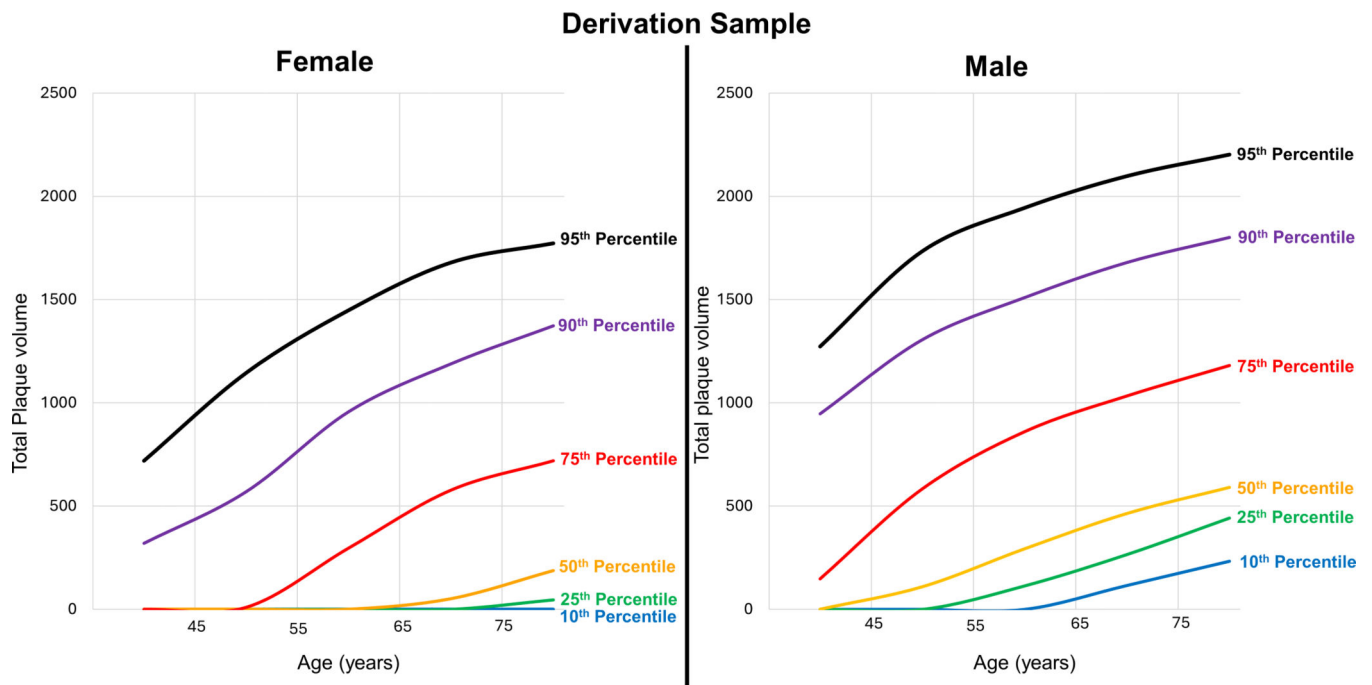
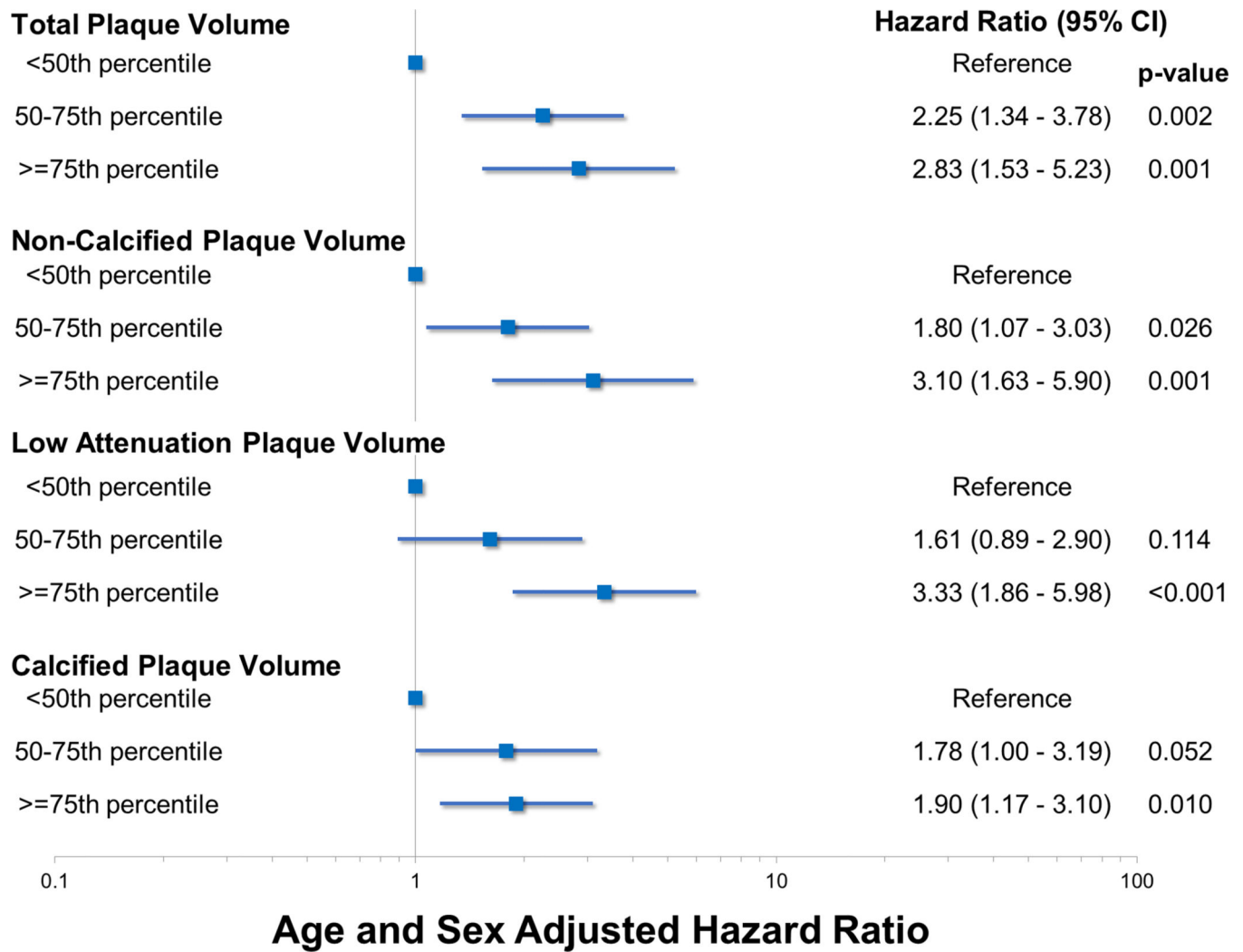


Figure 2: Distribution of total plaque volume as a function of sex and age. Expected plaque volume increase with age in female and male patients, with overall higher expected plaque volumes in male patients.

**Figure 3:**

Age and sex adjusted hazard ratio for fatal or non-fatal myocardial infarction in the combined external testing sample. CI: confidence interval.

Table 1

Characteristics for the derivation and two external testing samples

	Derivation Sample N=956	External SCOT- HEART N=1611	p-value vs. derivation	External St. Francis Hospital and Heart Center N=236	p-value vs. derivation
Age, median (IQR)	60 (56 – 70)	59 (52 – 65)	<0.001	65 (56 – 72)	0.140
Male, n (%)	707 (74.0)	928 (57.6)	<0.001	144 (60.5)	<0.001
Chest pain, n (%)	568 (59.4)	1,611 (100)	<0.001	125 (53.0)	0.077
Cardiovascular Risk Factors, n (%)					
Hypertension	521 (54.5)	576 (36.1)	<0.001	143 (60.1)	0.296
Diabetes	130 (13.6)	186 (11.6)	0.002	42 (17.7)	0.561
Dyslipidemia	668 (69.9)	804 (49.9)	<0.001	138 (58.0)	0.003
Smoking	86 (9.0)	298 (18.5)	<0.001	19 (8.0)	0.463
Family history of CAD	245 (25.6)	687 (43.0)	<0.001	68 (28.6)	0.160

CAD: coronary artery disease, IQR: interquartile range, SCOT-HEART: Scottish Computed Tomography of the Heart

Table 2

Multivariable models for the associations with fatal and non-fatal myocardial infarction

			Plaque Volume Percentile		Clinical and CAC Model		Stenosis Model		Plaque Model	
	Percentile	n	Unadjusted HR	p-value	Adjusted HR	p-value	Adjusted HR	p-value	Adjusted HR	p-value
Total plaque volume ^{*,†,‡}	<50 th	743	Reference	--	Reference	--	Reference	--	Reference	--
	50–75 th	466	2.25(1.35 – 3.75)	0.002	2.22(1.32 – 3.74)	0.003	1.17(1.27 – 3.70)	0.005	2.16(1.25 – 3.73)	0.005
	≥75 th	638	2.65(1.47 – 4.78)	0.001	2.50(1.29 – 4.86)	0.007	2.39(1.19 – 4.83)	0.015	2.38(1.06 – 5.33)	0.035
Non-calcified plaque volume ^{*,†,‡}	<50 th	732	Reference	--	Reference	--	Reference	--	Reference	--
	50–75 th	449	1.73(1.03 – 2.90)	0.037	1.77(1.04 – 2.99)	0.035	1.73(1.01 – 2.97)	0.046	1.75(1.01 – 3.03)	0.047
	≥75 th	666	3.04(1.62 – 5.68)	0.001	2.78(1.41 – 5.48)	0.003	2.66(1.28 – 5.53)	0.009	2.78(1.16 – 6.66)	0.022
Low-attenuation plaque volume ^{*,†,‡}	<50 th	719	Reference	--	Reference	--	Reference	--	Reference	--
	50–75 th	261	1.60(0.90 – 2.87)	0.111	1.60(0.88 – 2.90)	0.124	1.59(0.87 – 2.90)	0.130	1.61(0.88 – 2.95)	0.120
	≥75 th	867	3.02(1.74 – 5.22)	<0.001	3.12(1.69 – 5.74)	<0.001	3.07(1.60 – 5.90)	0.001	3.30(1.64 – 6.64)	0.001
Calcified plaque volume	<50 th	837	Reference	--	Reference	--	Reference	--	Reference	--
	50–75 th	304	1.71(0.96 – 3.05)	0.067	1.76(0.98 – 3.16)	0.059	1.68(0.93 – 3.05)	0.088	0.68(0.92 – 3.05)	0.089
	≥75 th	706	1.93(1.19 – 3.13)	0.007	1.72(1.02 – 2.91)	0.042	1.60(0.91 – 2.79)	0.100	1.63(0.93 – 2.87)	0.089
Non-calcified plaque burden	<50 th	992	Reference	--	Reference	--	Reference	--	Reference	--
	50–75 th	375	1.10(0.66 – 1.85)	0.714	1.17(0.68 – 2.01)	0.571	1.10(0.64 – 1.91)	0.724	0.74(0.38 – 1.46)	0.386
	≥75 th	480	1.32(0.73 – 2.37)	0.360	1.59(0.84 – 2.99)	0.152	1.43(0.74 – 2.77)	0.289	0.73(0.27 – 1.95)	0.532
Low-attenuation plaque burden ^{*,†,‡}	<50 th	687	Reference	--	Reference	--	Reference	--	Reference	--
	50–75 th	192	1.06(0.54 – 2.09)	0.858	0.88(0.44 – 1.77)	0.725	0.87(0.43 – 1.74)	0.691	0.89(0.44 – 1.80)	0.742
	≥75 th	968	2.10(1.27 – 3.48)	0.004	2.24(1.29 – 3.90)	0.004	2.13(1.20 – 3.78)	0.010	0.30(1.09 – 4.84)	0.029

* Multivariable analysis in the combined external sample. Each type of plaque volume or burden was assessed separately. The clinical and coronary artery calcium (CAC) model included age, sex, cardiovascular risk factors (hypertension, diabetes, dyslipidemia, smoking, family history), and CAC score. The stenosis model included the variables from the clinical and CAC model variables as well as obstructive coronary artery disease (defined as left main stenosis ≥50% or ≥70% in other segments). The plaque model included the variables from the stenosis model as well as plaque volume or burden as a continuous variable. Improvement in model fit by including plaque volume or burden percentile compared to the clinical and CAC model denoted as

‡ compared to the stenosis model as

†, and compared to the plaque model as